

Design of a Benchtop Power Supply

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Background:

This project's goal is to design and build a benchtop power supply containing an electronic circuit that provides an adjustable source of power for prototyping with microcontrollers and similar small electronics. While I tailored the power supply design to fit my needs, minor alterations could be made to fit other requirements. Emphasis was placed on the design of the user-interface and ease of manufacturing and assembly.

Requirements:

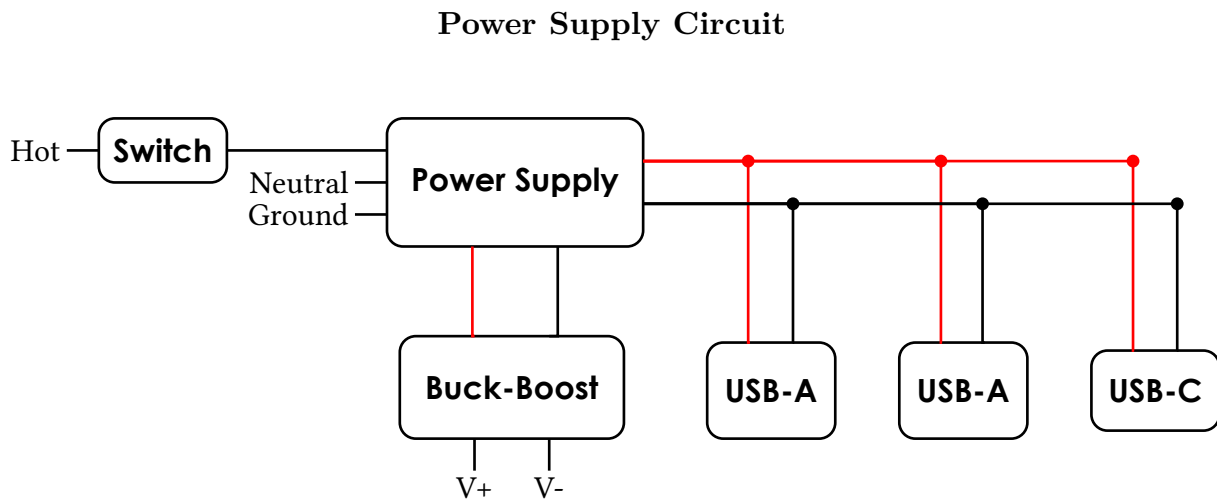
To determine the necessary components, functionality requirements were defined and basic calculations were performed to ensure all parts of the final system work harmoniously.

Functionality:

1. Adjustable voltage delivery for experimental circuits with over-current protection
2. USB-A and USB-C ports for powering microcontrollers
3. Compact desktop form factor
4. Intuitive and practical user-interface
5. Ease of assembly/disassembly

Circuit Design:

To enable the aforementioned functions, the following circuit was sketched. A mains switch is used to isolate the wall power from the circuit when the system is not in use. The power supply module outputs to both the adjustable buck-boost module and the USB modules.



Parts List & Brief Calculations:

Power Supply:

Assuming the highest possible load situation, the adjustable buck boost converter draws 40W, two USB-A ports draw 30W (15 each), and USB-C draws 65W. In total, the circuit draws 135W. A 150W supply would operate at 90% of its maximum capacity under this load, which provides adequate headroom. After selecting the size of the supply, the voltage and current must also be considered. There are many 150W power supply options including 5V, 12V, 24V, 36V, and 48V configurations, each supplying a current that upholds $P = IV$. The buck-boost module and USB buck converters support a range of input voltages, however, the USB-C board I chose only accepts an input voltage up to 30V. To reduce power loss and remain in the range of compatible voltages, I selected a 24V supply.

Adjustable Buck-Boost Module:

The specific buck-boost used in this project was chosen because of its output over-current protection, bright screen, and affordable price. The unit accepts 6-36V in and outputs adjustable voltage from 0.5-36V. Maximum output current depends on the voltage selected. From 0-8.8V, max current is 4.5A and from 8.8-36V, max current is defined by the curve: $I(v) = \frac{P}{V}$.

USB Modules & Miscellaneous Parts

For my use case, the USB-A and USB-C ports are nice additions to the core buck-boost functionality. I plan to power microcontrollers for testing and experimentation. However, a smaller, cheaper, and simpler adjustable power supply could be built without the 65W USB-C port. When searching for USB-C ports that support higher levels of power delivery, there are many standards of which to be mindful (such as USB PD).

Along with the larger parts, 18AWG wire, fork/spade connectors, a mains switch, and heat-shrink tubing were important electrical details.

Enclosure

I designed the enclosure for the electrical components to be manufactured using typical makerspace equipment. I planned on leveraging laser cutting and 3D printing for the enclosure panels. M3 hardware was used exclusively for assembly which further improves ease of construction and reduces the parts list.

Mechanical Design

I began the enclosure design by mocking up prototypes of the front panel layout using Figma (a 2D graphics software). These layouts helped narrow down the general form of the enclosure. After selecting two of my favorite front layouts, I moved to physical prototyping with cardboard. The cardboard prototyping phase proved helpful in refining my design selection. The prototype in [Figure: 4](#) was selected because of the minimal desk space required, easier design for manufacturing with $\frac{1}{8}$ in acrylic sheets, and preferable button layout.

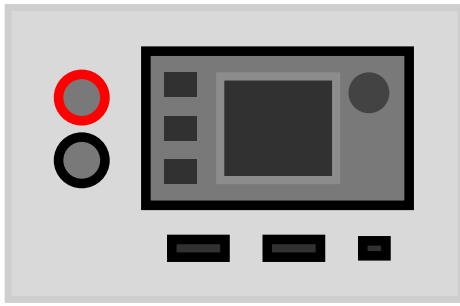


Figure 1: Horizontal front panel layout



Figure 2: Horizontal cardboard prototype

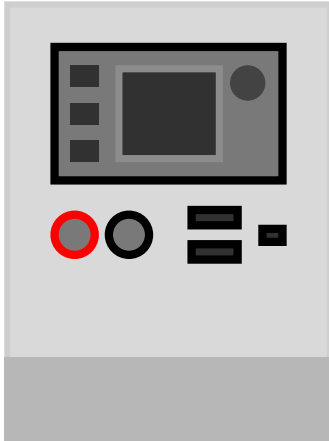


Figure 3: Vertical front panel layout



Figure 4: Vertical cardboard prototype

Mechanical Design Highlights

USB Modules – Panel Mounting: While mocking up the assembly in CAD, I realized the USB modules I planned on mounting behind the front panel would be difficult to secure. I wanted the modules to sit together and function as a sub-assembly of the front panel. They needed to be secure but allow for soldering access. This remains the weakest aspect of the design; it functions better than expected, but is far from elegant. I plan to redesign this area in the future.

Enclosure Size – Addition of Rear Vent: I wanted to shrink the size of the final housing, and since I decided to route the power cord out from the back panel (which meant the unit could not sit flush against a wall), I was able to rework the internal layout. I shifted the PSU module rearward and designed a vent that bumps out on the rear panel to accommodate the new PSU position. I designed the vent to be a removable access point for some of the PSU's electrical connections. The removable design also made the 3D printing process easier by allowing for fewer supports and less overall material. The addition of the access point also improved the assembly process as there are many connections routed into and out of the PSU module.

Manufacturing & Assembly

3D printing the front and rear panels required multiple attempts, as warping on the print bed caused parts to fall outside of critical tolerances.

The enclosure was designed to use exclusively M3 hardware, and notably, no adhesive was required for assembly.

Photos

Here are a few photos of the completed design:



Figure 5: Power supply $\frac{3}{4}$ view.



Figure 6: Power supply front view.



Figure 7: Power supply rear view.

Final Thoughts

As I use this benchtop power supply, I plan to continue to iterate the front panel interface/layout. In addition, I plan to continue to iterate on the PSU circuit. In the near future I am looking to add a cooling fan triggered by a thermistor to improve safety and component lifetime. And, eventually, I would love to slowly replace the PSU module with a self-designed circuit that would include a transformer, diode bridge, filtering capacitors, and voltage regulators.

This project is available on GitHub; if you are interested in building your own, my STL and DXF files are open source for 3D printing and laser cutting, respectively.

Appendix

Bill of Materials

Quantity				Description	Cost
1				Benchtop Power Supply Assembly	\$66.84
.	1			Front Panel Assembly	\$19.22
.	.	1		Front Panel	\$0.336
.	.	1		Buck Boost Module	\$10.3
.	.	1		USB-A Assembly	\$3.19
.	.	.	2	USB-A Module	\$2.88
.	.	.	1	Upper USB Bracket	\$0.024
.	.	.	1	Middle USB Bracket	\$0.024
.	.	.	1	Lower USB Bracket	\$0.024
.	.	.	4	M3x10mm Button Head Bolt - Bracket	\$0.16
.	.	.	2	M3x6mm Button Head Bolt - Mounting	\$0.08
.	.	1		USB-C Assembly	\$2.67
.	.	.	1	USB-C Module	\$2.57
.	.	.	1	USB-C Bracket	\$0.06
.	.	.	1	M3x6mm Button Head Bolt	\$0.04
.	.	1		Banana Jack - Red	\$1.36
.	.	1		Banana Jack - Black	\$1.36
.	1			Rear Panel Assembly	\$2.84
.	.	1		Rear Panel	\$0.36
.	.	1		AC Switch	\$0.71
.	.	1		AC Receptacle - IEC 320-C14	\$1.06
.	.	1		Vent	\$0.07
.	.	16		M3x6mm Button Head Bolt	\$0.64
.	1			Power Circuit	\$36.18
.	.	1		135W SMPS Power Supply	\$21.56
.	.	6		Spade Terminal Connector	\$5.79
.	.	6		18AWG Wire	\$8.33
.	.	1		Heat Shrink Tubing	\$0.5
.	1			Right Acrylic Panel	\$2.15
.	1			Left Acrylic Panel	\$2.15
.	1			Top Acrylic Panel	\$2.15
.	1			Bottom Acrylic Panel	\$2.15